Raj Reddy

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Education

University of Utah — **Salt Lake City, UT**
B.S. Computer Science — *August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies — **Dallas, TX**
Associate Software Engineer — *June 2025 – Present*

- Developed real-time signal processing algorithms in Python and C++ on embedded systems, applying DSP techniques for defense sensor data streams.
- Built microservices for high-frequency data ingestion and storage, leveraging Podman and serverless infrastructure on AWS.
- Automated CI/CD pipelines with static code analysis and integration testing.

Doxy.me — **Charleston, SC**
Software Engineer — *August 2024 – May 2025*

- Built AI data platforms and internal tooling in Python and TypeScript, orchestrating LLM-powered inference pipelines (PyTorch, Llama) that processed high-volume clinical data from WebRTC sessions and surfaced structured outputs to internal and external-facing dashboards on AWS SageMaker and Fargate.
- Improved data quality and reliability by designing SQL-backed data models and REST APIs that normalized inputs from disparate clinical systems into consistent schemas, enabling accurate downstream analytics and reporting for clinical and operational teams.
- Identified and drove platform improvements end-to-end without a ticket queue, owning architecture decisions, CI/CD configuration, and observability tooling on AWS to keep systems reliable as the product scaled.

University of Utah — **Salt Lake City, UT**
Undergraduate Research Assistant – FuTURES Lab — *May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform *(Class Best):* Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.